Ashwin Satvi

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Education

Veermata Jijabai Technological Institute (VJTI) **Sep 2024 – Present**
B.Tech in Computer Engineering *Mumbai, Maharashtra*

M.K. Junior College **Jul 2021 – Mar 2023**
Higher Secondary Certificate (HSC) *Palghar, Maharashtra*

BMT English Medium High School **Jun 2020 – Apr 2021**
Secondary School Certificate (SSC) *Palghar, Maharashtra*

Relevant Coursework

- Data Structures and Algorithms
- Design and Analysis of Algorithms
- Operating Systems
- Database Management Systems
- Object-Oriented Programming
- Discrete Mathematics
- Machine Learning
- Computer Architecture
- Python Development
- Web Development
- Linear Algebra
- Probability and Statistics

Projects

StackForge – Cloud Native DevOps Platform | GitHub **Aug 2026 – Present**

- Built a full-stack internal developer platform using Next.js, TypeScript, FastAPI, PostgreSQL, and Redis for project, deployment, and pipeline management.
- Implemented JWT-based authentication and PostgreSQL-backed user management with a full REST API for project and deployment CRUD operations.
- Built an interactive deployment workflow (Source, Build, Test, Security Scan, Deploy, Verify) with history and status tracking.
- Containerized all services (frontend, backend, PostgreSQL, Redis, worker) using Docker Compose for reproducible development.

Real-Time Object Detection and Tracking System **Feb 2026 – Present**

- Developed a real-time computer vision system for multi-object detection and tracking from live video streams using Python and OpenCV.
- Integrated YOLO for object detection and DeepSORT/SORT to maintain object identities across frames for surveillance and traffic monitoring.
- Optimized inference performance through efficient frame processing and model tuning.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript
Frontend: HTML, CSS, React.js, Next.js, Tailwind CSS
Backend: FastAPI, Flask, REST APIs, SQLAlchemy
Databases: MySQL, PostgreSQL, Redis
DevOps & Systems: Docker, Docker Compose, Linux
CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming
Tools: Git, GitHub, VS Code, Kaggle, Google Colab
Machine Learning: Scikit-learn, Pandas, NumPy, Matplotlib